

---

This is an electronic reprint of the original article.  
This reprint may differ from the original in pagination and typographic detail.

Le Boulout, Yohann; Ehnholm, Gösta; Nieminen, Heikki  
**Multi-modal transducer-waveguide construct coupled to a medical needle**

*Published in:*  
Journal of the Acoustical Society of America

*DOI:*  
[10.1121/10.0022326](https://doi.org/10.1121/10.0022326)

Published: 22/11/2023

*Document Version*  
Publisher's PDF, also known as Version of record

*Published under the following license:*  
CC BY

*Please cite the original version:*  
Le Boulout, Y., Ehnholm, G., & Nieminen, H. (2023). Multi-modal transducer-waveguide construct coupled to a medical needle. *Journal of the Acoustical Society of America*, 154, 3388-3396.  
<https://doi.org/10.1121/10.0022326>

---

This material is protected by copyright and other intellectual property rights, and duplication or sale of all or part of any of the repository collections is not permitted, except that material may be duplicated by you for your research use or educational purposes in electronic or print form. You must obtain permission for any other use. Electronic or print copies may not be offered, whether for sale or otherwise to anyone who is not an authorised user.

NOVEMBER 22 2023

## Multi-modal transducer-waveguide construct coupled to a medical needle

Yohann Le Boulout  ; Gösta Ehnholm; Heikki J. Nieminen 

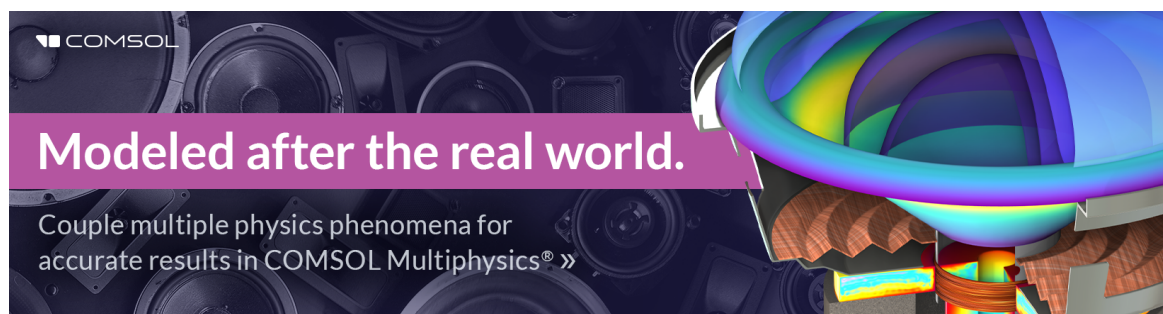


*J. Acoust. Soc. Am.* 154, 3388–3396 (2023)

<https://doi.org/10.1121/10.0022326>



CrossMark



## Multi-modal transducer-waveguide construct coupled to a medical needle

Yohann Le Boulout,  Gösta Ehnholm, and Heikki J. Nieminen<sup>a)</sup> 

Medical Ultrasonics Laboratory (MEDUSA), Department of Neuroscience and Biomedical Engineering, Aalto University, Rakentajanaukio 2, Espoo, 02150, Finland

### ABSTRACT:

Annually, more than  $16 \times 10^9$  medical needles are consumed worldwide. However, the functions of the medical needle are still limited mainly to cutting and delivering material to or from a target site. Ultrasound combined with a hypodermic needle could add value to many medical applications, for example, by reducing the penetration force needed during the intervention, adding precision by limiting the needle deflection upon insertion into soft tissues, and even improving tissue collection in fine-needle biopsy applications. In this study, we develop a waveguide construct able to operate a longitudinal-flexural conversion of a wave when transmitted from a Langevin transducer to a conventional medical needle, while maintaining high electric-to-acoustic power efficiency. The optimization of the waveguide structure was realized *in silico* using the finite element method followed by prototyping the construct and characterizing it experimentally. The experiments conducted at low electrical power consumption (under 5 W) show a 30 kHz flexural needle tip displacement up to 200  $\mu\text{m}$  and 73% electric-to-acoustic power efficiency. This, associated with a small sized transducer, could facilitate the design of ultrasonic medical needles, enabling portability, batteryization, and improved electrical safety, for applications such as biopsy, drug and gene delivery, and minimally invasive interventions. © 2023 Author(s). All article content, except where otherwise noted, is licensed under a Creative Commons Attribution (CC BY) license (<http://creativecommons.org/licenses/by/4.0/>).

<https://doi.org/10.1121/10.0022326>

(Received 25 March 2023; revised 29 September 2023; accepted 16 October 2023; published online 22 November 2023)

[Editor: Juan Tu]

Pages: 3388–3396

### I. INTRODUCTION

The hypodermic needle is one of the most commonly used instruments in healthcare with  $16 \times 10^9$  needles consumed annually.<sup>1,2</sup> They are frequently used as standard procedure for tissue sampling<sup>3,4</sup> or injection of fluids.<sup>5–9</sup> However, only limited development has taken place in their design for several decades.<sup>10,11</sup> This has caused complications in some applications, for example, targeting a specific area for anesthesia or targeting tumor in biopsy<sup>12</sup> [up to 34% inadequacy rate in breast biopsy using Fine Needle Aspiration (FNA)<sup>13</sup> and up to 21% in thyroid].<sup>14</sup> Moreover, approximately 10% of the population avoid medical treatment due to fear of needles,<sup>15</sup> potentially influencing access to healthcare.<sup>16</sup> Therefore, considerable improvements to medical needles are required.

In FNA, the insertion and guidance of the needle are facilitated by its back-and-forth translation,<sup>17</sup> rotation around the center axis, and tilting.<sup>18</sup> At a much faster rate than possibly produced by a human operator, an alternative way consists of generating ultrasonic mechanical waves, resulting in microscale motions of the needle tip.<sup>19</sup>

Actuation of the needle tip by ultrasound has recently been demonstrated to be useful for many medical purposes: to decrease the penetration force<sup>20–23</sup> for an easier and spatially more accurate procedure, provide improved visibility

of the biopsy needle under ultrasound Doppler imaging,<sup>24–26</sup> and hypothetically causing less pain by reducing the penetration resistance.<sup>20,27–29</sup>

We have recently developed an ultrasound-enhanced fine needle aspiration biopsy (USEFNAB) method, which allows the introduction of flexural ultrasound waves to a medical needle and needle tip to improve the tissue yield in biopsy.<sup>19</sup> In USEFNAB, a piezoelectric actuator is employed to produce kHz longitudinal waves, which are converted into flexural waves, essentially producing mechanical movement of the needle tip laterally with respect to the needle center axis. However, optimization in power-efficiency has not been yet presented. Clinically, a small and fully portable device would typically be considered preferable for better usability, the potential to batteryize, and patient or operator safety, which could be enabled by improved electrical-to-acoustical power-conversion efficiency.

In this study, we aimed to develop a new acoustic waveguide construct with the intent to optimize the electrical-to-acoustical power-conversion efficiency and be capable of converting longitudinal to flexural mode, followed by validation of the technical performance.

### II. MATERIAL AND METHODS

Using the finite element (FE) method, we chose to first model the Langevin transducer that was selected as the

<sup>a)</sup>Email: [heikki.j.nieminen@aalto.fi](mailto:heikki.j.nieminen@aalto.fi)

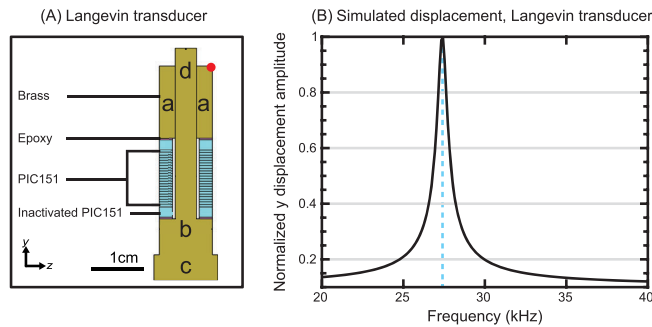


FIG. 1. (Color online) (A) Cross-section of the Langevin transducer construct with 22 rings arranged so that their electric fields direction through the piezo were opposite with adjacent piezo elements. At each end of the piezo stack, an extra inactivated piezo element was placed to isolate the transducer from the rest of the system for safety. At the extremities of the transducer, two masses made of brass were included (a, b, and c). An M4 bolt (d) connecting the two masses was applied to pre-stress the piezo stack with 3.1 kN. (B) Displacement measured from a frequency domain simulation using COMSOL Multiphysics 6.1 in the y-direction at the red dot. The blue dotted line indicated the resonance frequency of the Langevin transducer.

sound source (Fig. 1). This was followed by stepwise optimization of the geometry of the full system comprising a piezoelectric transducer, a waveguide, and a biopsy needle. Based on the simulations, we then built custom-made

prototypes, followed by characterizing their performance and comparing the performance with simulations.

Two waveguide constructs were studied: (i) a system with a transducer center axis placed perpendicular to that of the needle [Fig. 2(A)] and (ii) a system with a transducer center axis aligned at an offset and parallel to that of the needle [Fig. 2(B)], the needle is a standard hypodermic needle with a lancet tip (21 gauge, i.e., 21 G:  $\phi_{in} \times \phi_{out} = 0.514 \times 0.8$  mm, length 120 mm, model: 466564/3, 100 Sterican, B. Braun Medical, Inc., Melsungen, Germany). The purpose of the waveguide is to convert the longitudinal wave mode in the transducer to a flexural wave mode in the needle, as well as to match the mechanical impedance of the transducer to the mechanical impedance of the needle. The appropriate mass distribution, shape, and positioning of the structures in relation to electrical input are critical to achieving optimum performance of the system. Therefore, to understand the wave propagation and optimize the system, an acoustic simulation was conducted *in silico* employing a FE method (COMSOL Multiphysics 6.1, Inc., Burlington, MA). In the following, the steps realized during this optimization are elaborated.

### A. Numerical simulations

For optimization of the arrangement, the FE method was used. First, consider the piezoelectric phenomenon,

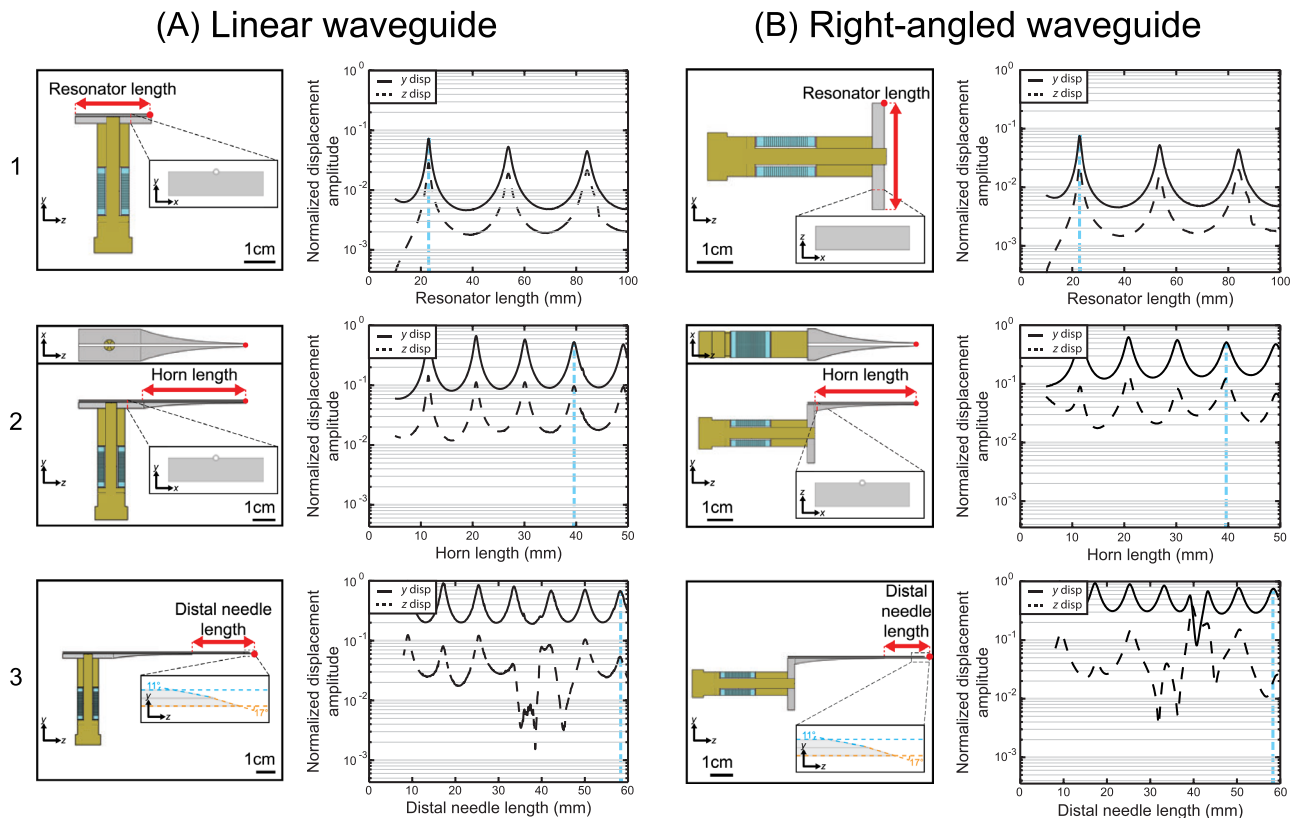


FIG. 2. (Color online) Numerically generated modal responses of the intermediate designs for both the linear (A) and the right-angled (B) waveguide constructs, whether constituted of the Langevin transducer and (1) a resonator, (2) a resonator and a horn, or (3) a resonator, a horn and a distal needle. For the purpose of amplitude optimization, the normalized displacement amplitudes in both the y- and z-directions are displayed as a function of the length of (1) the resonator, (2) the exponentially tapered horn, and (3) the distal needle. In each case (1–3), the chosen length of the structure is indicated on the graph by the blue dotted line, and the flexural displacement amplitude in the y-dimension corresponds to the one measured at the red dot on the associated schematics.

which consists of creating electric potential over the material if the material is under stress (direct electric piezo effect) or the generation of mechanical strain when an electrical field is applied on the material (reverse piezo electric effect).<sup>30,31</sup> In our model, the stress-charge constitutive relation has been used,<sup>32</sup>

$$\begin{cases} \mathbf{T} = c_E \mathbf{S} - e^t \mathbf{E}, \\ \mathbf{D} = e \mathbf{S} + \epsilon_S \mathbf{E}, \end{cases} \quad (1)$$

where  $\mathbf{T}$  is the stress vector (Pa),  $\mathbf{S}$  is the strain vector (unitless),  $\mathbf{E}$  is the electrical field vector ( $\text{V m}^{-1}$ ),  $\mathbf{D}$  is the vector of electric displacement field ( $\text{C m}^{-2}$ ),  $c_E$  is the elasticity matrix (Pa),  $e$  is the piezoelectric constant ( $\text{C m}^{-2}$ ),  $e^t$  is the transposed piezoelectric constant ( $\text{C m}^{-2}$ ), and finally  $\epsilon_S$  is the dielectric permittivity at constant strain ( $\text{F m}^{-1}$ ).

To study the vibration throughout the acoustic part of the device, the elastic wave equation, obtained from Newton's second law, is then solved in a frequency domain perturbation study:

$$-\rho \omega^2 \mathbf{u} = \nabla \cdot (FS)^T + \mathbf{F}_V e^{i\phi}, \quad (2)$$

where  $\rho$  is the density of the material ( $\text{kg m}^{-3}$ ),  $\omega$  is the complex angular frequency ( $\text{rad s}^{-1}$ ),  $\mathbf{u}$  is the displacement (m), and  $FS$  is the first Piola-Kirchhoff stress ( $\text{N m}^{-2}$ ), which when transposed is the nominal stress.  $\mathbf{F}_V$  is the volume force ( $\text{N m}^{-3}$ ) and  $e^{i\phi}$  refers to the alternating electrical field with phase  $\phi$  applied to the piezo elements.

The simulations were conducted in two steps, a stationary study was performed to calculate the deformation of the structure induced by the bolt pretension over the piezo stack using the following equation [Eq. (3)]:

$$0 = \nabla \cdot \mathbf{S} + \mathbf{F}_V, \quad (3)$$

where  $\mathbf{S}$  is the second Piola-Kirchhoff stress ( $\text{N m}^{-2}$ ), followed by solving the elastic wave equation, Eq. (2).

To avoid infinite vibration amplitude, when the system is in resonance, an artificial damping ratio, i.e., *Rayleigh* damping, was set to 0.001 (unitless). The different parts of the model were meshed using the swept technique for the piezo stack and the free tetrahedral technique for the rest of the geometry. The mesh size was implemented to include at least 20 elements per wavelength.

The parameters of the different materials used in the models are given in the Table I.

## B. Simulation of the Langevin transducer

The modelled Langevin sandwich transducer, based on a commercially available piezo stack [P-010.10H, Physik Instrumente (PI) GmbH & Co. KG, Karlsruhe, Germany], was composed of 22 ring-type piezo elements (PIC151; dimensions =  $\phi_{\text{out}} \times \phi_{\text{in}} \times \text{thickness} = 10 \times 5 \times 0.5 \text{ mm}$ ) stacked and fastened by a bolt to achieve a pretension force (3.1 kN). Each element was arranged so that the direction of the electric fields through the piezo was opposite as compared to the adjacent elements. Electrically, the elements were all connected in parallel: In this way, the required driving amplitude voltage was decreased by a factor of 22 compared to using a single element having a thickness of the full stack. At each end of the stack, an extra inactivated piezo element (PIC151, dimensions =  $\phi_{\text{out}} \times \phi_{\text{in}} \times \text{thickness} = 10 \times 5 \times 1 \text{ mm}$ ) was added to electrically isolate the transducer from the rest of the system for safety. To simulate the glue used by the manufacturer to hold the stack together, two layers of epoxy were added (epoxy, dimensions =  $\phi \times H = 10 \times 0.2 \text{ mm}$ ) at each end of the stack. Two masses of brass, custom-built, were added on each side of the piezo stack connected with the bolt, as detailed in the following, to shift the resonance frequency of the piezo stack to the targeted resonance of the entire construct (approximately 30 kHz). The front mass of the transducer [Fig. 1(A.a)] was cylindrical (brass, dimensions =  $H \times \phi = 14 \times 10 \text{ mm}$ ). The back mass [Figs. 1(A.b) and 1(A.c)] was shaped to facilitate the fixation of the stack inside the enclosure. An M4 screw [Fig. 1(A.d)] (brass,  $H \times \phi = 35 \times 4 \text{ mm}$ ) was connected to a cylindrical front mass (brass,  $H \times \phi = 3 \times 10 \text{ mm}$ ) and to a cube-shaped back mass [Fig. 1(A.b)] (brass,  $H \times W \times L = 10 \times 10 \times 10 \text{ mm}$ ) connected to a rectangular cuboid [Fig. 1(A.c)] (brass,  $H \times W \times L = 5 \times 10 \times 12 \text{ mm}$ ). The M4 screw, cylindrical, cube, and rectangular cuboid were all machined from one piece.

To fasten all the elements mentioned previously to create a stack, the M4 bolt linked the back mass and the top mass in order to apply the pretension force (3.1 kN) over the stack. The bolt connects both ends through the top mass via a thread and exposes 3 mm of the bolt, which was used to connect the Langevin transducer to the waveguide, using the waveguide's threaded hole. Finally, after a pre-study calculating the bolt pretension effect over the system, a frequency domain with perturbation simulation from 20 to 40 kHz was conducted to find the resonance frequency of the system.

TABLE I. Material parameters used for the FE simulations.

| Material                  | Density [ $\text{kg/m}^3$ ] | Young's modulus [GPa] | Poisson's ratio | Isotropy    | Reference                                 |
|---------------------------|-----------------------------|-----------------------|-----------------|-------------|-------------------------------------------|
| Brass                     | 8750                        | 115                   | 0.307           | Isotropic   | Ref. 33                                   |
| Epoxy                     | 1250                        | 3.5                   | 0.3             | Isotropic   | Ref. 34                                   |
| PIC151                    | 7760                        | Table II              | 0.34            | Anisotropic | Physik Instrumente datasheet <sup>a</sup> |
| EOS Stainless steel 316 L | 8000                        | 193                   | 0.265           | Isotropic   | Ref. 33                                   |
| AISI 304                  | 8070                        | 205                   | 0.275           | Isotropic   | Ref. 35                                   |

<sup>a</sup>Data provided upon request by Physik Instrumente.

### C. Simulations of the waveguides

Connecting the needle directly to the transducer without impedance matching would lead to compromised transmission of acoustic power from the transducer to the needle. The transducer stack is resonating in a longitudinal mode with a relatively small amplitude and has mechanically a high impedance. On the other hand, the needle should vibrate with a large amplitude, approximately  $200\ \mu\text{m}$  perpendicularly to its center axis. The needle is flexurally relatively compliant, thereby having a low mechanical impedance. Therefore, a mechanical impedance matching from the transducer to the needle is needed, which was implemented with a resonator (Fig. 2.1) connected to a horn (Fig. 2.2). In addition, the connection point of the transducer to a resonator converts wave modes from a longitudinal to a flexural mode at the needle.

We investigated two waveguide geometries: a linear waveguide and a right-angled waveguide [Figs. 2(A) and 2(B)].

The linear waveguide was made of EOS Stainless steel 316L (surgical quality) and included a rectangular resonator in the proximal end [dimensions =  $L \times W \times H$  = variable length (from 8 to 100 mm with 0.1 mm steps)  $\times 12 \times 3$  mm] (Fig. 2.1). From here, the waveguide extrudes as a tapered section named horn, with the width (W) and the height (H) decreasing in width and height exponentially along the length towards the distal end (Fig. 2.2) [ $L$  = variable length (from 5 to 50 mm with 0.1 mm increments), dimensions starting point =  $W \times H = 12 \times 3$  mm, dimensions ending point =  $W \times H = 0.9 \times 0.5$  mm] (Fig. 2.2). The tapering amplifies the acoustic signal and facilitates mechanical impedance matching: The constant flow of power will propagate as a wave with increasing amplitude into more flexible regions at smaller cross-sections of the horn while minimizing reflections of energy. The waveguide also included a groove along the length of the waveguide to provide a coupling area for the needle cannula (dimensions =  $\phi_{\text{in}} \times \phi_{\text{out}} = 0.514 \times 0.8$  mm).

The second design, right-angled waveguide, included a linear portion with a  $90^\circ$  angle implemented in the beginning of the horn [Fig. 2(B)]. The rationale for this geometry is to improve the ergonomics by aligning the center axes of the transducer and the needle to be parallel.

We conducted the waveguide optimization in two steps: First, the length of the resonator was varied and simultaneously the absolute displacement obtained at one end of the resonator was recorded (Fig. 2.1, red dots). The smallest length of the resonator with a local maximum displacement was selected as the horn length for the next step. Second, the length of the horn was varied, and simultaneously the absolute displacement obtained at the distal end of the horn was recorded. The length of the horn was selected by choosing a length that provided a local maximum in the displacement at the tip of the horn (Fig. 2.2, red dots), still being sufficiently short to enable usable operation (subjective evaluation). The frequency used in optimizing resonator and horn lengths was 27.4 kHz, which is the natural resonance frequency of the transducer obtained from the transducer simulation study (Fig. 1).

### D. Simulation of the distal needle length

After optimization of the waveguide, the distal part of the needle was included in the simulation to understand its effect on the transducer and find an optimal position. The distance from the tip of the waveguide (horn part) to the needle tip was varied from 8 to 60 mm at  $100\ \mu\text{m}$  increments without load constraint at the tip. The simulated needle follows the specification of a standard 21 G hypodermic needle (dimensions =  $\phi_{\text{in}} \times \phi_{\text{out}} = 0.514 \times 0.8$  mm) made of AISI 304. The needle tip, based on standard hypodermic needle tip, is composed of two bevels which have opening angles of  $17^\circ$  and  $11^\circ$ , respectively, observed as a side projection to yz plane (Fig. 2.3). Similar to the waveguide optimization, the needle length was varied and simultaneously the absolute displacement obtained at the needle tip was recorded (Fig. 2.3, red dots). The distal needle length was selected by choosing a length that provided a local maximum in the displacement at the tip of the needle, allowing at least 50 mm penetration depth inside tissue as measured from the site of insertion.

### E. Impedance characteristics of the experimental system and simulation

After building the experimental systems, the agreement between the experimental system and the simulation was verified in terms of impedance characteristics as follows: The Langevin transducer was coupled to the waveguide, which was a three-dimensionally (3D) printed EOS stainless steel 316L object (3D Formtech Oy, Jyväskylä, Finland). The needle (21 G  $\times$  120 mm, model: 466564/3, 100 STERICAN, B Braun, Melsungen, Germany), placed in the groove, was brazed onto the last 20 mm of the waveguide's distal end.

The simulated electrical impedance of the arrangement was obtained from COMSOL using a frequency domain with perturbation simulation (20–40 kHz with 50 Hz steps). The impedance measurement was conducted using a Digilent Analog Discovery 2 combined with the WaveForms software (Digilent, Inc., Henley Court Pullman, WA).

### F. Quantifying the needle tip displacement

To observe the needle tip during sonication we used a high-speed camera (model: Phantom V1612, Vision Research, Inc., Wayne, NJ) in conjunction with a macro lens (model: Canon MP-E 65 mm f / 2.8 1–5x Macro Photo, Canon, Inc., Tokyo, Japan). The camera was placed in front of an acrylic chamber (external dimensions =  $L \times W \times H = 21 \times 14 \times 15$  mm, wall thickness = 5 mm) filled with air or de-ionized water degassed to 6 mg/L at ambient temperature ( $22.3^\circ\text{C}$ ). The needle was immersed to a depth of 30 mm from the water surface.

The system was driven with an incoming signal generated by a Digilent Analog Discovery 2 at the resonance frequency ( $f \approx 30\text{--}34$  kHz), which was defined by the maximum power transmission (difference of the incident

and reflected powers), for ten cycles. This was connected to a custom-made class B amplifier with an output impedance of 50 Ohms coupled with a custom-made step-up transformer (11:20 turn ratio) delivering an incoming instantaneous power of 5 W (schematics of the setup are shown in Fig. 3).

High-speed camera videos were taken in order to acquire the needle tip displacement during sonication [sample rate =  $10 \times$  the driving frequency (frames per second), exposure time =  $0.8 \mu\text{s}$ , resolution  $128 \times 128$  pixels, angular magnification =  $5\times$ , pixel size =  $5.6 \mu\text{m}$ ]. The videos were then analyzed in MATLAB (Release 2020b, The MathWorks, Inc., Natick, MA) using an algorithm capable of tracking the needle position using cross correlation frequency technique to allow subpixel image registration<sup>36</sup> based on the work of Perra *et al.*<sup>19</sup> Translating the needle of a range of  $50 \mu\text{m}$  at  $1 \mu\text{m}$  steps with a high-precision stage (Z825B stage, Thorlabs, Inc., Newton, NJ; stage precision: 29 nm), the true and optically measured displacement correlated with Pearson correlation coefficient of 0.9982 with a slope of 1.005. A total of ten repetitions per variation were realized.

### G. Calorimetry

To define the total acoustic power emitted from the needle, a calorimetric assessment was done. The calorimeter (material = thermal wall insulator, wall thickness = 3 cm) was filled with 25 ml of de-ionized water (at room temperature:  $22^\circ\text{C}$ ) mixed with a magnetic stirrer (C-MAG HS 4, IKA-Werke GmbH & Co. KG, Staufen, Germany). The needle tip was then immersed to a depth of 30 mm from the water surface.

The system was operating at resonance frequency ( $f \approx 30.5\text{--}32.0 \text{ kHz}$ ), which was defined as the frequency

providing the maximum power transmission (difference of the incident and reflected burst-averaged electrical powers) within the frequency range 30–34 kHz (calibration). The signal used during sonication was composed of bursts (succession of activated signals over a certain period of time followed by a non-activated signal) of 300 cycles with a 55% duty cycle during 30 s. The incident and reflected powers were measured and time-averaged over a burst period: The consumed electrical energy is equal to their difference.<sup>37</sup> Different consumed electrical energy were tested, from 0 to 3 J at 0.3 J steps. The water temperature change was recorded using a PT 3000 thermometer probe for 60 s before and after the sonication. After averaging the temperature data, the difference of the two temperatures gave the heat deposited into a known volume of distilled water for a certain incident energy. The calculation of the heat deposition was obtained from the Eq. (4),

$$P_{\text{heat}} = \frac{m \times C \times \Delta T}{t}, \quad (4)$$

where  $m$  is the water mass (g),  $C$  is the specific heat capacity of water [ $\text{J}/(\text{g } ^\circ\text{C})$ ],  $\Delta T$  the difference of the average temperature before and after sonication ( $^\circ\text{C}$ ), and  $t$  is the sonication time (s). The efficiency of the system was defined by the slope of the line obtained from a linear fit on the measurements of energy deposition as a function of consumed electrical energy.

## III. RESULTS

### A. Langevin transducer

The first analysis aimed to define the resonance frequency of the transducer. The computational results indicate a peak vibration amplitude around 27.4 kHz (vibration of  $2.9 \mu\text{m}$  at 5 W input electrical power), as shown in Fig. 1.

### B. Optimization of the resonators, horns, and needle length

The purpose of this simulation was to find a geometry of the waveguide, which would provide a great displacement in  $y$  direction while avoiding “spurious” displacement in other directions due to different modes. In Fig. 2(A.1), there is a clear recurrence of the greater displacement values at every multiple of a flexural wavelength (23 mm) in the resonator (stainless steel, thickness: 3 mm) at 27.4 kHz. For practical reasons, the length of the resonator was selected to be the shortest possible i.e., 23 mm. Maximum displacement at the tip of the horn [Fig. 2(A.2)], occurred at every multiple of half of the flexural wavelength (stainless steel). However, the wavelength was shorter because the wavelength depends on the structure thickness,<sup>38</sup> which decreases exponentially along the horn (varying thickness: from 3 to 1 mm).

To minimize the reflection of the flexural wave in the tapering part of the horn, one should select a long horn; however, the longer the horn is, the more compromised the usability is (maneuvering by the operator), thus 39.5 mm was selected. The displacement is greatly

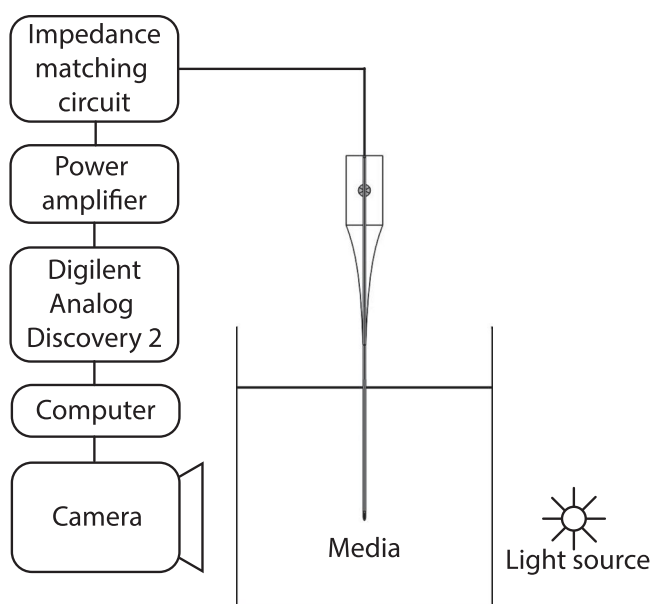


FIG. 3. Schematics of the experiment arrangement to optically measure the ultrasonic needle tip activity.

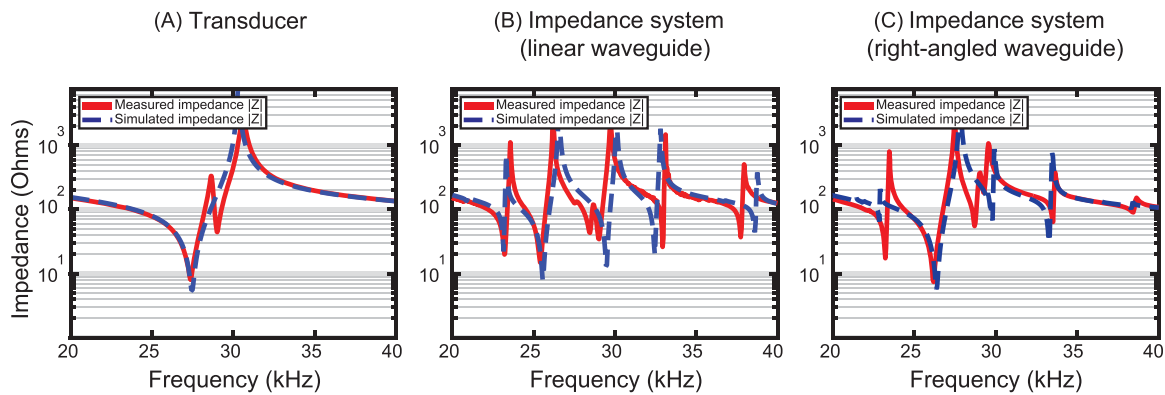


FIG. 4. (Color online) (A) Comparison between the simulated impedance of the transducer construct and the measured impedance of the custom-designed transducer. (B) Comparison of the whole system impedance including the transducer, linear waveguide and needle simulation, and experimental device with linear waveguide. (C) Comparison of the whole system with right-angled waveguide and needle simulation and experimental device with right-angled waveguide.

amplified by the tapered shape of the horn [Fig. 2(A.2), flexural displacement of  $48.6\ \mu\text{m}$  at 5 W electrical input power] compared to that of the resonator structure [Fig. 2(A.1), flexural displacement of  $4.8\ \mu\text{m}$ ]. The needle displacement [Fig. 2(A.3)] is also greater than the horn construct (flexural displacement of  $88\ \mu\text{m}$ ), this is due to the tapered distal end of the needle, which behaves as an amplifier. From the data set, it seems that every half flexural wavelength provides a local maximum displacement ( $\lambda \approx 16.6\ \text{mm}$ ) in AISI 304 material (cross-sectional shape is a ring:  $\phi_{\text{out}} = 0.8\ \text{mm}$   $\phi_{\text{in}} = 0.514\ \text{mm}$ ). The selected distance distal part of the needle to distal part of the horn was selected to be 58 mm, which was selected for practical reasons considering the biopsy applications.

The results shown in Fig. 2(B), using the right-angled waveguide, suggest a similar functioning as the linear waveguide. Similar dimensions were selected as 23 mm for the resonator (generating flexural displacement of  $4.5\ \mu\text{m}$ ), 39.5 mm for the horn (flexural displacement of  $35\ \mu\text{m}$ ), and 58 mm for the distal length of the needle (flexural displacement of  $117\ \mu\text{m}$ ).

### C. Validation model—System

To compare the difference between the simulations and our prototypes, an impedance measurement with a bare stack transducer and the whole systems were compared with the respective simulated impedances in the frequency range 20–40 kHz (Fig. 4). The simulation curve and the measured peaks exhibit similar behavior. The main impedance peak of the transducer is at 30.6 kHz, while with the linear and right-angled waveguide system, the impedance peak is measured at 29.7 and 27.5 kHz, respectively.

### D. Experimental arrangement

The different manufactured prototypes are demonstrated in Fig. 5. Comparison of the waveguide structures are shown in Figs. 5 and 6 demonstrating that both of the designs can create flexural waves at the needle tip.

Simulations at 5 W input power showed a displacement peak of  $88\ \mu\text{m}$  at the needle tip for the linear waveguide [Fig. 5(A)] and  $117\ \mu\text{m}$  for the right-angled waveguide [Fig. 5(B)]. Simulations suggested similar displacements

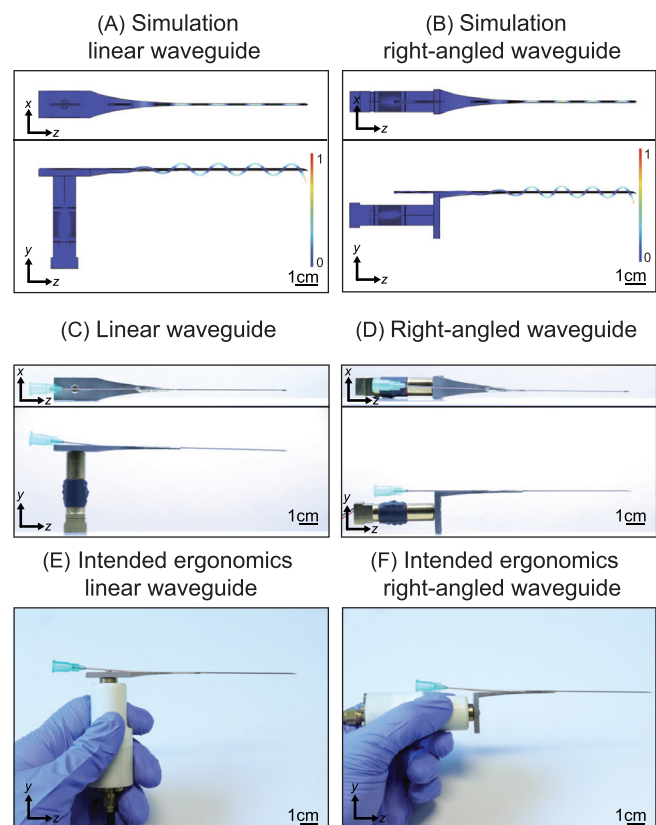
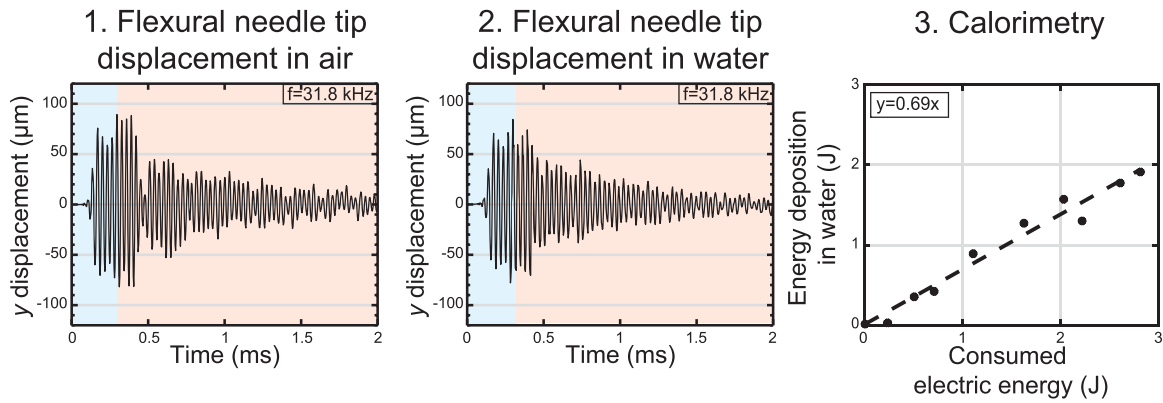


FIG. 5. (Color online) Visuals of the device with the waveguides. Examples of displacements obtained from the simulation of the two waveguides, i.e., linear (A) and right-angled (B). Photographs of the waveguides connected to the Langevin transducer (C and D) and their intended handling with the housing (E and F). The frequency domain simulation of the linear (A) and right-angled (B) waveguides demonstrate the displacements in the geometry and have been exaggerated solely for better visualization of the sub-millimeter displacements.

## (A) Linear waveguide



## (B) Right-angled waveguide

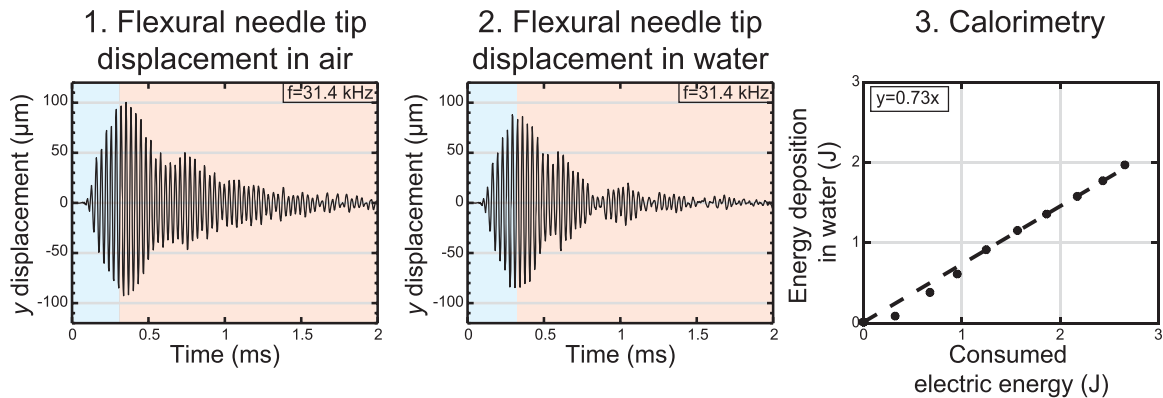


FIG. 6. (Color online) Experimental displacement of the needle using the linear (A) and right-angled waveguide (B) in air (1) or water (2) ( $f \approx 31.4$ – $31.8$  kHz, ten cycles, instantaneous electrical power 5 W, in blue background) recorded with a high-speed camera. (3) Displays their needle tip heating deposition in water.

than experiments: The measurements at 5 W in air [Figs. 6(A.1) and 6(B.1)] recorded  $89 \pm 2 \mu\text{m}$  (mean  $\pm$  standard deviation, SD;  $n = 10$ ) and  $102 \pm 1 \mu\text{m}$  ( $n = 10$ ) for the linear and right-angled waveguide, respectively. In water, the displacements at the needle tip were up to  $87 \pm 6 \mu\text{m}$  (mean  $\pm$  SD;  $n = 10$ ) and  $92 \pm 1 \mu\text{m}$  ( $n = 10$ ), for the linear and right-angled waveguide, respectively. The coefficient of variation percentage (CV%) for displacements were in the range 1.5%–6.5%, suggesting acceptable repeatability. Moreover, it seems that a slightly greater electrical-to-acoustical power-efficiency of 73% was obtained with the right-angled waveguide compared to 69% with the linear waveguide [Figs. 6(A.3) and 6(B.3)].

## IV. DISCUSSION

The results of the *in silico* structural optimization demonstrated the displacements at the needle tip were significantly dependent on the dimensions of the waveguide design and needle-to-waveguide coupling point, the greatest displacements were obtained when the system was in resonance. In the simulations, an increase in displacement was

observed at the end of each stage, for example, resonator, horn, or needle, respectively. Importantly, the experimentally demonstrated power-efficiency was good, 69% with the linear waveguide, 73% with the right-angled waveguide, which is in line with the literature.<sup>37,39</sup> Several factors may contribute to the power-efficiency, as outlined in the following.

The extrusion of the linear rectangular segment of the resonator extends symmetrically from the transducer coupling point towards opposite directions. Symmetry minimizes the generation of bending modes within the transducer, which, if present, could reduce electro-acoustic efficiency of the transducer. Bending waves in the transducer would yield tension and compression in different parts of the piezo stack simultaneously, compromising the performance, which is now eliminated. Furthermore, the exponential tapered shape of the horn gives an amplification of the signal by having approximately the same energy in the smaller cross section, while minimizing the reflection from the boundaries towards the source. The geometry, therefore, provides geometric amplification at the same time with successful impedance matching from the higher mechanical

impedance of the transducer to the lower mechanical impedance of the needle. The needle bevel provides further an additional state of displacement amplification towards the tip of the needle, where the action of the ultrasound is intended.<sup>19</sup>

The studied waveguides, together with the transducer alignment, successfully converted the longitudinal waves of the transducer to a flexural displacement of the needle tip. The experiments with needle in air provided large peak-to-peak displacement amplitudes up to 200 and 180  $\mu\text{m}$  in water. The practical relevance of the potential to generate large displacements is selecting the relevant magnitude needed for a specific application in question. For example, gentle actuation of tissue by the needle or adjacent micro-bubbles could be used to improve tissue yield in a biopsy application, whereas generation of large amplitudes and violent cavitation clouds could be useful in an intervention application such as the histotripsy of a tumor. Several reports have shown that the actuation of the needle tip by ultrasound might be also useful for medical purposes such as improving spatial accuracy of needle insertion<sup>20</sup> and avoiding tissue deformation<sup>21</sup> or hypothetically pain reduction.<sup>19,20,27–29,40</sup>

None of the studies mentioned before reported on the optimization of electro-acoustic efficiency. Better efficiency could contribute to lower operation voltages enabling better patient safety and could further enable, for example, batterization and miniaturization if developed to the full potential. The instantaneous electrical input power into the transducer was below 5 W and the displacements demonstrated were sufficient required for the biopsy application.<sup>19</sup> Compared to the previous research device used by Perra *et al.*,<sup>19</sup> a hand-piece volume and a mass decrease of 70% and 34%, respectively, have been achieved with the new design. Feedback from medical doctors confirmed the clinically acceptable usability of the miniaturized ultrasonic devices. However, the right-angled waveguide construct was preferred over the linear waveguide for its “pen-like” ergonomics [Fig. 5(F)] resembling needle devices currently used clinically.

The loading of the needle is expected to affect the resonance frequency; therefore, the system could in the future be improved with resonance frequency tracking, which was not implemented in this study and is a limitation. In our experiment, the effect of the water on the frequency change was not investigated, but stiffer and more viscous media could have a greater effect on the frequency, leading to a reduction in the efficiency.

## V. CONCLUSION

Ultrasound in medicine is a well-known technology and is widely used for different kinds of therapeutic procedures. However, flexurally oscillating needles have received limited attention in the literature. Since the needle is a common tool in healthcare but with clear limitations, this paper showed a novel way of improving needles. By actuating the needle with flexural waves, we may be able to decrease the penetration force needed during the insertion of the needle

or improve tissue extraction during biopsies. Because of the efficiency of the system, the design could lead to better patient safety of flexurally oscillating medical needles, and enable batterization and miniaturization of the system, which are key factors influencing user adaptation to new ultrasonic technologies in the medical setting. Therefore, this study opens insights towards developing flexurally oscillating ultrasound-actuated medical needles for a variety of medical purposes including biopsy, drug and gene delivery, and minimally invasive interventions.

## ACKNOWLEDGMENT

We thank Emanuele Perra, D.Sc., Dr. Kristofer Nyman, MD, Ph.D., Dr. Minna Rehell, MD, Dr. Jyrki Kreula, MD, Ph.D., Dr. Kenneth P.H. Pritzker, MD, FRCPC, Matti Mikkola, M.Sc., Saif Bunni, M.Sc., Jouni Rantanen, M.Sc., Maxime Fauconnier, Ph.D. and Dr. Nick Hayward, MBBS, Ph.D. for constructive discussions related to the topic. Business Finland and Academy of Finland are acknowledged for financial support (Business Finland: Grant No. 5607/31/2018; Academy of Finland: Grant Nos. 314286, 311586, and 335799). H.J.N. and Y.L.B. have stock ownership and stock options, respectively, in Swan Cytologics, Inc., Toronto, ON, Canada. H.J.N. is an inventor within the patent family associated with WO2018000102A1. Y.L.B., G.E. and H.J.N. are inventors with the patent family associated with WO2020240084A1. Y.L.B. and G.E. are inventors with the patent family associated with EP4101402A1. All authors contributed to the design of the study, writing or reviewing the manuscript and have approved the submitted version of the manuscript. Y.L.B. produced the data and conducted the data analysis. G.E. assembled the transducer construct and created the electronics for the system. Curated data are available upon request.

## APPENDIX

See Table II for material data set.

TABLE II. Material data set PIC151, from the manufacturer.

| Coefficient | unit                       | Value                    | Coefficient | unit                       | Value                     |
|-------------|----------------------------|--------------------------|-------------|----------------------------|---------------------------|
| s11E        | $\text{m}^2 \text{N}^{-1}$ | $1.683 \times 10^{-11}$  | s11D        | $\text{m}^2 \text{N}^{-1}$ | $1.436 \times 10^{-11}$   |
| s33E        | $\text{m}^2 \text{N}^{-1}$ | $1.900 \times 10^{-11}$  | s33D        | $\text{m}^2 \text{N}^{-1}$ | $9.750 \times 10^{-11}$   |
| s55E        | $\text{m}^2 \text{N}^{-1}$ | $5.096 \times 10^{-11}$  | s55D        | $\text{m}^2 \text{N}^{-1}$ | $2.924 \times 10^{-11}$   |
| s12E        | $\text{m}^2 \text{N}^{-1}$ | $-5.656 \times 10^{-12}$ | s12D        | $\text{m}^2 \text{N}^{-1}$ | $-8.112 \times 10^{-12}$  |
| s13E        | $\text{m}^2 \text{N}^{-1}$ | $-7.107 \times 10^{-12}$ | s13D        | $\text{m}^2 \text{N}^{-1}$ | $-2.2507 \times 10^{-12}$ |
| s44E        | $\text{m}^2 \text{N}^{-1}$ | $5.096 \times 10^{-11}$  | s44D        | $\text{m}^2 \text{N}^{-1}$ | $2.924 \times 10^{-11}$   |
| s11E        | $\text{m}^2 \text{N}^{-1}$ | $4.497 \times 10^{-11}$  | s11D        | $\text{m}^2 \text{N}^{-1}$ | $4.497 \times 10^{-11}$   |
| c11E        | $\text{N m}^{-2}$          | $1.076 \times 10^{+11}$  | c11D        | $\text{N m}^{-2}$          | $1.183 \times 10^{+11}$   |
| c33E        | $\text{N m}^{-2}$          | $1.004 \times 10^{+11}$  | c33D        | $\text{N m}^{-2}$          | $1.392 \times 10^{+11}$   |
| c55E        | $\text{N m}^{-2}$          | $1.962 \times 10^{+11}$  | c55D        | $\text{N m}^{-2}$          | $3.420 \times 10^{+10}$   |
| c12E        | $\text{N m}^{-2}$          | $6.312 \times 10^{+10}$  | c12D        | $\text{N m}^{-2}$          | $7.376 \times 10^{+10}$   |
| c13E        | $\text{N m}^{-2}$          | $6.385 \times 10^{+10}$  | c13D        | $\text{N m}^{-2}$          | $4.436 \times 10^{+10}$   |
| c44E        | $\text{N m}^{-2}$          | $1.962 \times 10^{+10}$  | c44D        | $\text{N m}^{-2}$          | $3.420 \times 10^{+10}$   |
| c66E        | $\text{N m}^{-2}$          | $2.224 \times 10^{+10}$  | c66D        | $\text{N m}^{-2}$          | $2.224 \times 10^{+10}$   |

- <sup>1</sup>A. M. Hauri, G. L. Armstrong, and Y. J. Hutin, "The global burden of disease attributable to contaminated injections given in health care settings," *Int. J. Std. Aids* **15**(1), 7–16 (2004).
- <sup>2</sup>Y. J. Hutin, A. M. Hauri, and G. L. Armstrong, "Use of injections in healthcare settings worldwide, 2000: Literature review and regional estimates," *BMJ* **327**(7423), 1075 (2003).
- <sup>3</sup>R. G. Amedee and N. R. Dhurandhar, "Fine-needle aspiration biopsy," *Laryngoscope* **111**(9), 1551–1557 (2001).
- <sup>4</sup>A. A. Renshaw and N. Pinnar, "Comparison of thyroid fine-needle aspiration and core needle biopsy," *Am. J. Clin. Pathol.* **128**(3), 370–374 (2007).
- <sup>5</sup>D. A. Cardone and A. F. Tallia, "Joint and soft tissue injection," *Am. Family Phys.* **66**(2), 283–288 (2002).
- <sup>6</sup>D. Wynaden, I. Landsborough, S. McGowan, Z. Baigomohamad, M. Finn, and D. Pennebaker, "Best practice guidelines for the administration of intramuscular injections in the mental health setting," *Int. J. Mental Health Nursing* **15**(3), 195–200 (2006).
- <sup>7</sup>L. P. Aiello, A. J. Brucker, S. Chang, E. T. Cunningham, D. J. D'Amico, H. W. Flynn, L. R. Grillone, S. Hutcherson, J. M. Liebmann, T. P. O'Brien, I. U. Scott, R. F. Spaide, C. Ta, and M. T. Trese, "Evolving guidelines for intravitreal injections," *Retina (Philadelphia, PA)* **24**, S3–S19 (2004).
- <sup>8</sup>L. Guarda-Nardini, D. Manfredini, and G. Ferronato, "Arthrocentesis of the temporomandibular joint: A proposal for a single-needle technique," *Oral Surg. Oral Med. Oral Pathol. Oral Radiol. Endodontol.* **106**(4), 483–486 (2008).
- <sup>9</sup>J. R. Lenz, D. D. Degnan, J. B. Hertig, and J. G. Stevenson, "A review of best practices for intravenous push medication administration," *J. Infusion Nursing* **40**(6), 354–358 (2017).
- <sup>10</sup>J. Kirkup, "Surgical history. The history and evolution of surgical instruments. VIII. Catheters, hollow needles and other tubular instruments," *Ann. R. College Surgeons Engl.* **80**(2), 81–90 (1998).
- <sup>11</sup>C. Ball and R. Westhorpe, "Local anaesthesia-the continuing evolution of spinal needles," *Anaesthesia Intensive Care* **32**(1), 3 (2004).
- <sup>12</sup>K. P. Pritzker and H. J. Nieminen, "Needle biopsy adequacy in the era of precision medicine and value-based health care," *Arch. Pathol. Lab. Med.* **143**(11), 1399–1415 (2019).
- <sup>13</sup>E. D. Pisano, L. L. Fajardo, J. Tsimikas, N. Sneige, W. J. Frable, C. A. Gatsonis, W. P. Evans, I. Tocino, and B. J. McNeil, "Rate of insufficient samples for fine-needle aspiration for nonpalpable breast lesions in a multicenter clinical trial: The Radiologic Diagnostic Oncology Group 5 Study," *Cancer: Interdiscip. Int. J. Am. Cancer Soc.* **82**(4), 679–688 (1998).
- <sup>14</sup>H. Gharib, J. R. Goellner, and D. A. Johnson, "Fine-needle aspiration cytology of the thyroid: A 12-year experience with 11,000 biopsies," *Clin. Lab. Med.* **13**(3), 699–709 (1993).
- <sup>15</sup>Y. Nir, A. Paz, E. Sabo, and I. Potasman, "Fear of injections in young adults: Prevalence and associations," *Am. J. Trop. Med. Hygiene* **68**(3), 341–344 (2003).
- <sup>16</sup>J. G. Hamilton, "Needle phobia: A neglected diagnosis," *J. Family Practice* **41**(2), 169–182 (1995).
- <sup>17</sup>J. Kreula, P. Virkkunen, and S. Bondestam, "Effect of suction on specimen size in fine-needle aspiration biopsy," *Investigative Radiol.* **25**(11), 1175–1181 (1990).
- <sup>18</sup>P. Han and K. Ehmann, "Study of the effect of cannula rotation on tissue cutting for needle biopsy," *Med. Eng. Phys.* **35**(11), 1584–1590 (2013).
- <sup>19</sup>E. Perra, E. Lampisjärvi, G. Barreto, M. Arif, T. Puranen, E. Haeggström, K. P. Pritzker, and H. J. Nieminen, "Ultrasonic actuation of a fine-needle improves biopsy yield," *Sci. Rep.* **11**(1), 1–15 (2021).
- <sup>20</sup>X. Liao, M. Sadiq, G. Corner, S. Cochran, and Z. Huang, "Reduced penetration force through ultrasound activation of a standard needle: An experimental and computational study," in *Proceedings of the IEEE International Ultrasonics Symposium*, Prague, Czech Republic (July 21–25, 2013), pp. 1436–1439.
- <sup>21</sup>X. Liao, M. Sadiq, Y. Kuang, G. Corner, S. Cochran, and Z. Huang, "Performance optimization of ultrasonic needle actuating device for insertion operation into tissue mimics," in *Proceedings of the IEEE International Ultrasonics Symposium*, Chicago, IL (September 3–6, 2014), pp. 823–826.
- <sup>22</sup>L. Gui, R. Xiao, X. Liao, Y. Kuang, M. Sadiq, Z. Chunnian, S. Cochran, and Z. Huang, "Automatic frequency tracking system for needle actuating device," in *Proceedings of the IEEE International Ultrasonics Symposium*, Chicago, IL (September 3–6, 2014), pp. 815–818.
- <sup>23</sup>M. Yang and J. D. Zahn, "Microneedle insertion force reduction using vibratory actuation," *Biomed. Microdev.* **6**(3), 177–182 (2004).
- <sup>24</sup>Y. Kuang, A. Hilgers, M. Sadiq, S. Cochran, G. Corner, and Z. Huang, "Modelling and characterisation of a ultrasound-actuated needle for improved visibility in ultrasound-guided regional anaesthesia and tissue biopsy," *Ultrasonics* **69**, 38–46 (2016).
- <sup>25</sup>M. R. Sadiq, S. Cochran, X. Liao, and Z. Huang, "Enhanced US-guided needle intervention through ultrasound actuation of a standard needle," in *Proceedings of the IEEE International Ultrasonics Symposium*, Chicago, IL (September 3–6, 2014), pp. 827–830.
- <sup>26</sup>T. Jiang, G. McLeod, Z. Huang, X. Zhu, Y. Jiao, X. Li, Z. Shen, and Y. Cui, "Localization accuracy of ultrasound-actuated needle with color Doppler imaging," *Diagnostics* **10**(12), 1020 (2020).
- <sup>27</sup>R. Lehtinen and E. Oksala, "Penetration of disposable needles," *Int. J. Oral Surg.* **8**(2), 145–148 (1979).
- <sup>28</sup>H. Egekvist, P. Bjerring, and L. Arendt-Nielsen, "Regional variations in pain to controlled mechanical skin traumas from automatic needle insertions and relations to ultrasonography," *Skin Res. Technol.* **5**(4), 247–254 (1999).
- <sup>29</sup>H. Egekvist, P. Bjerring, and L. Arendt-Nielsen, "Pain and mechanical injury of human skin following needle insertions," *Eur. J. Pain* **3**(1), 41–49 (1999).
- <sup>30</sup>S. O. R. Moheimani and A. J. Fleming, *Piezoelectric Transducers for Vibration Control and Damping* (Springer-Verlag, Berlin, 2006).
- <sup>31</sup>Z. Wei and L. K. Weavers, "Combining COMSOL modeling with acoustic pressure maps to design sono-reactors," *Ultrasonics Sonochem.* **31**, 490–498 (2016).
- <sup>32</sup>A. Meitzler, H. Tiersten, A. Warner, D. Berlincourt, G. Couquin, and F. Welsh III, "IEEE standard on piezoelectricity," <https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=26560> (Last viewed January 18, 2023).
- <sup>33</sup>D. O. Kipp, *MatWeb Metal Material Data Sheets (MDS)* (MatWeb, LLC., Blacksburg, VA, 2021).
- <sup>34</sup>Society for the Advancement of Material and Process Engineering, "SAMPE—Conference and Exhibition, May 20–23, 2019—Conference/ May 21–22, 2019—Exhibition, Charlotte Convention Center/Charlotte, North Carolina," <https://app.knovel.com/hotlink/itble/rcid:kpSAMPECE8/id:kpSAMPECE8/sampe-conference-exhibition/sampe-conference-exhibition> (Last viewed August 30, 2022).
- <sup>35</sup>Battelle Memorial Institute, *Metallic Materials Properties Development and Standardization (MMPDS-11)* (Battelle Memorial Institute, Columbus, OH, 2016).
- <sup>36</sup>M. Guizar-Sicairos, S. T. Thurman, and J. R. Fienup, "Efficient subpixel image registration algorithms," *Opt. Lett.* **33**(2), 156–158 (2008).
- <sup>37</sup>F. Li, C. Chen, W. Li, and D. Zeng, "The electro-acoustic output behavior and thermal stability of 1–3 piezoelectric composite transducers applied to FUS surgery," *J. Mater. Sci.: Mater. Electron.* **31**(15), 12066–12073 (2020).
- <sup>38</sup>H. N. Abramson, H. J. Plass, and E. A. Ripperger, "Stress wave propagation in rods and beams," *Adv. Appl. Mech.* **5**, 111–194 (1958).
- <sup>39</sup>S. Lin and F. Zhang, "Measurement of ultrasonic power and electro-acoustic efficiency of high power transducers," *Ultrasonics* **37**(8), 549–554 (2000).
- <sup>40</sup>K. K. Kundan, S. Laha, and A. Ghatak, "Vibration assisted puncturing of a soft brittle solid," *Extreme Mech. Lett.* **26**, 26–34 (2019).